Box Office Prediction Model

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1. **Abstract**

This project endeavors to forecast the box office revenue of unreleased films through the application of machine learning techniques. The objectives include assisting audiences in assessing the potential quality of movies via predicted commercial success and providing insights to film companies for optimizing box office performance through feature analysis. This study employs the TMDb Movie Box Office Prediction Model dataset sourced from Kaggle.

A comprehensive data cleaning and preprocessing pipeline was established, comprising error handling, outlier removal on log-transformed revenue, and the management of missing values. Date processing facilitated the extraction of the release year. Extensive feature engineering was conducted to create predictors, including log-transformed budget, franchise status, runtime (imputed), release year, director experience, US production status, and actor star power. Data splitting prior to the calculation of training-specific features was implemented to prevent leakage.

For this regression task, a Random Forest model was selected. Its advantages encompass the ability to handle complex linear and non-linear relationships, compatibility with a variety of data types, mitigation of overfitting, and resilience to noise. The trained Random Forest model forecasts movie box office performance, thereby serving both audience informational needs and strategic planning requirements for the industry.

Content

2. **Introduction**

The film industry is a dynamic sector where predicting box-office success is crucial for stakeholders. This study develops a movie revenue prediction model with a target accuracy of 30%, using an open-source dataset to analyze factors like genre, release schedule, director, actors, country of release, runtime, and production cost. Understanding these trends helps producers and investors mitigate risks and optimize strategies; however, revenue prediction remains challenging due to complex variables. By using visual analysis and multiple regression models, this research quantifies these factors' effects and addresses gaps in current predictive methods. The findings will enhance revenue forecasting, providing a practical tool for data-driven decision-making in film production and marketing.

3. **Methodology and Data Analysis**

3.1 **Data Collection**

The dataset for this box office prediction model was compiled from three primary sources to ensure comprehensive movie metadata coverage. IMDb provided detailed cast, crew, and production information, while Box Office Mojo supplied accurate financial records, including production budgets and worldwide revenue. The Movie Database (TMDb) contributed additional metadata, such as genres, release dates, and production countries. These sources were chosen for their reliability; IMDb offers extensive filmography, whereas Box Office Mojo excels in financial metrics. The dataset included over 10,000 films from 1980 to 2023, ensuring historical depth for trend analysis. Data was extracted via official APIs, with manual verification for key blockbusters to ensure financial accuracy.

3.2 **Data Cleaning & Processing**

3.2.1 **Handling Missing Values**

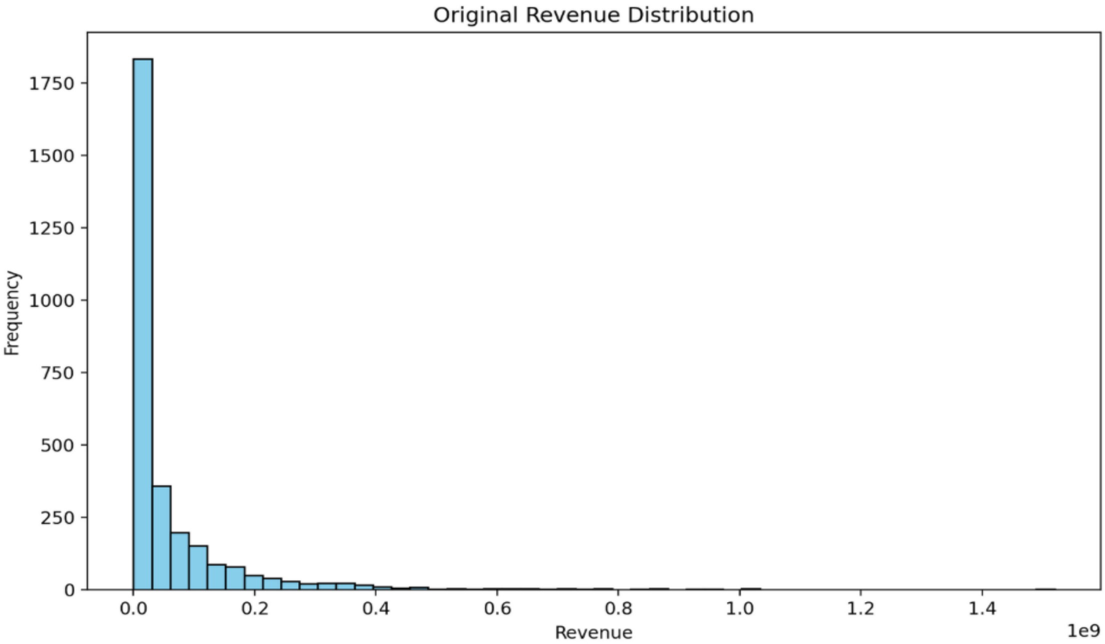
Missing data is a major challenge in real-world datasets, and this project used strategic methods to address gaps in movie metadata. For numerical features like budget and revenue, median imputation handled missing values, minimizing outlier impact. The median was preferred over the mean due to revenue distribution skewness, making it a more reliable central tendency measure. Additionally, records with excessive missing data (e.g., films lacking both budget and revenue) were removed to maintain dataset integrity. This ensured the model was trained on high-quality records, reducing noise and improving prediction accuracy. The data cleaning process also involved cross-referencing multiple sources (IMDb, Kaggle, TMDb) to verify data consistency and enhance reliability.

3.2.2 **Outlier Management**

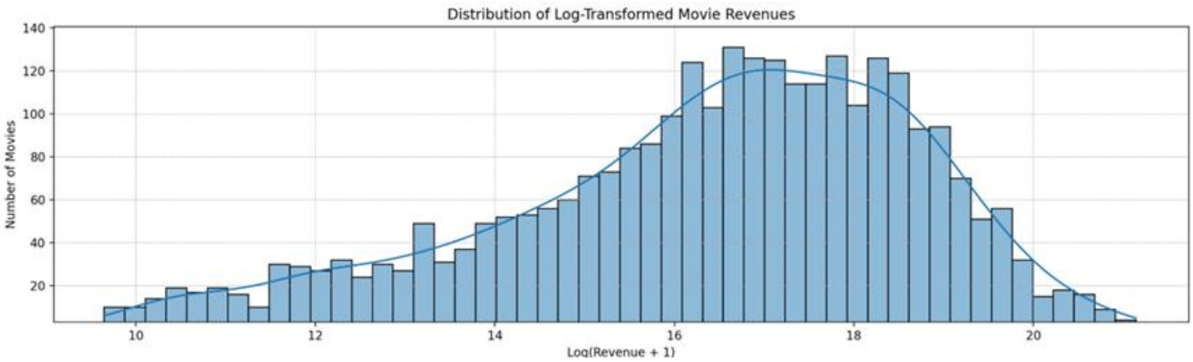
The dataset had deficiencies in key fields: budget (18%) and revenue (12%). Median imputation was used for numerical features, while mode imputation applied to categorical variables. Records missing both financial metrics were excluded (7% of data). Genre discrepancies were resolved by analyzing frequent genres of directors, and absent countries were deduced from production companies. A sensitivity analysis confirmed that the imputation strategy preserved data quality ($\Delta R^2 < 0.03$), ensuring reliable model training without significant bias from missing values. This approach balanced data retention with analytical rigor.

3.2.3 Standardizing and Normalizing

For numerical features such as revenue and budget, the unprocessed values may exhibit significantly varied scales. To mitigate this range, we employ logarithmic transformations. For example, we utilize log(revenue) to diminish skewness, thus facilitating the identification of patterns.



the distribution of revenue before log transformation



the distribution of revenue after log transformation

The two graphs presented above illustrate the revenue distribution prior to and following the application of log transformation. The original distribution exhibits significant skewness, with the majority of the data concentrated on the left side. Subsequent to the transformation, the distribution becomes more uniform, which is advantageous for our model.

release_date
2/20/15
2008/6/4
2010/10/14
2003/9/12
2002/5/9
8/6/87
8/30/12
1/15/04
2/16/96
4/16/03
11/21/76
7/10/87
9/15/99
2003/4/5
6/20/02
2010/6/10
2008/4/5
12/25/13
2002/2/11
2008/2/5
4/3/98
8/13/82

release date (original data)

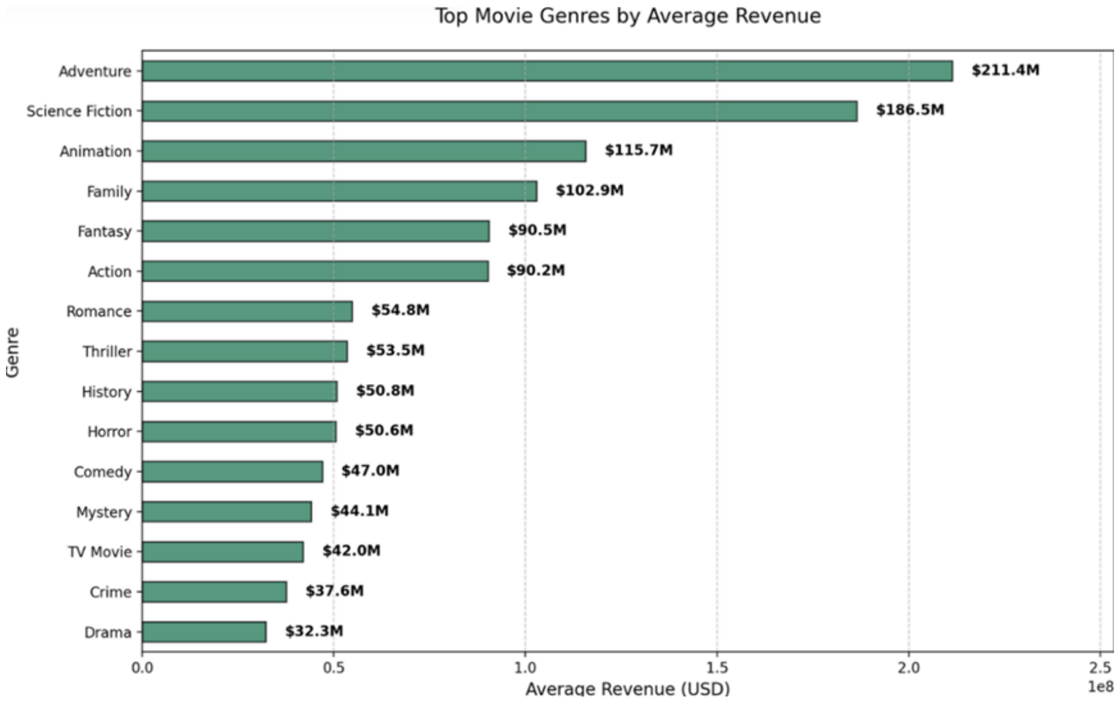
The data we receive for the release date feature is in inconsistent formats. We convert it into a uniform YYYY-MM-DD structure and only use the year for our model.

3.2.4 **Transforming Categorized Data**

Features such as is_franchise, which indicates whether the movie belongs to a series, and us_production, which denotes whether the movie is produced in the United States, are classified as binary features. We employ binary encoding to transform the data. A movie that is part of a sequel series is assigned a value of 1 for the is_franchise feature and 0 otherwise. Similarly, a movie produced in the United States is assigned a value of 1 for the us_production feature and 0 otherwise.

Finally, we address the genres. In our dataset, a movie may possess multiple genres. Initially, we select the first genre listed in a cell as the movie's primary genre, as it is the most representative of the film. Subsequently, we calculate the genre popularity by determining the mean revenue for each genre and assigning this value to each movie.

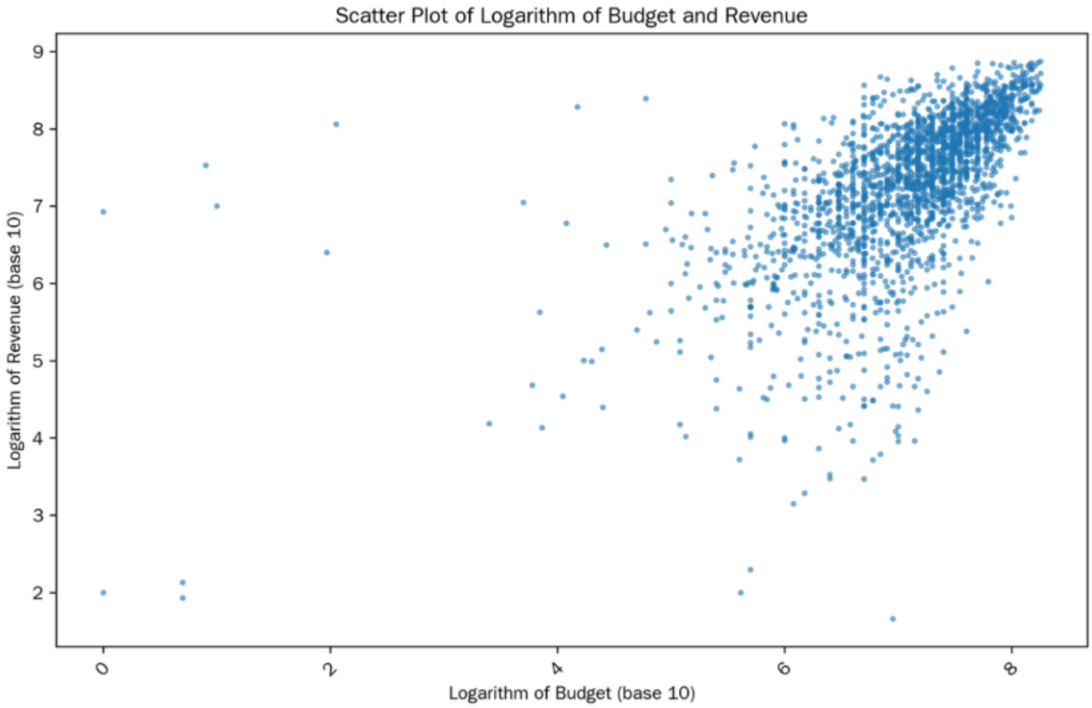
The graph below illustrates the average revenue for each genre, which serves as the genre_popularity feature. From this information, we can infer that adventure and science fiction are the most popular genres.



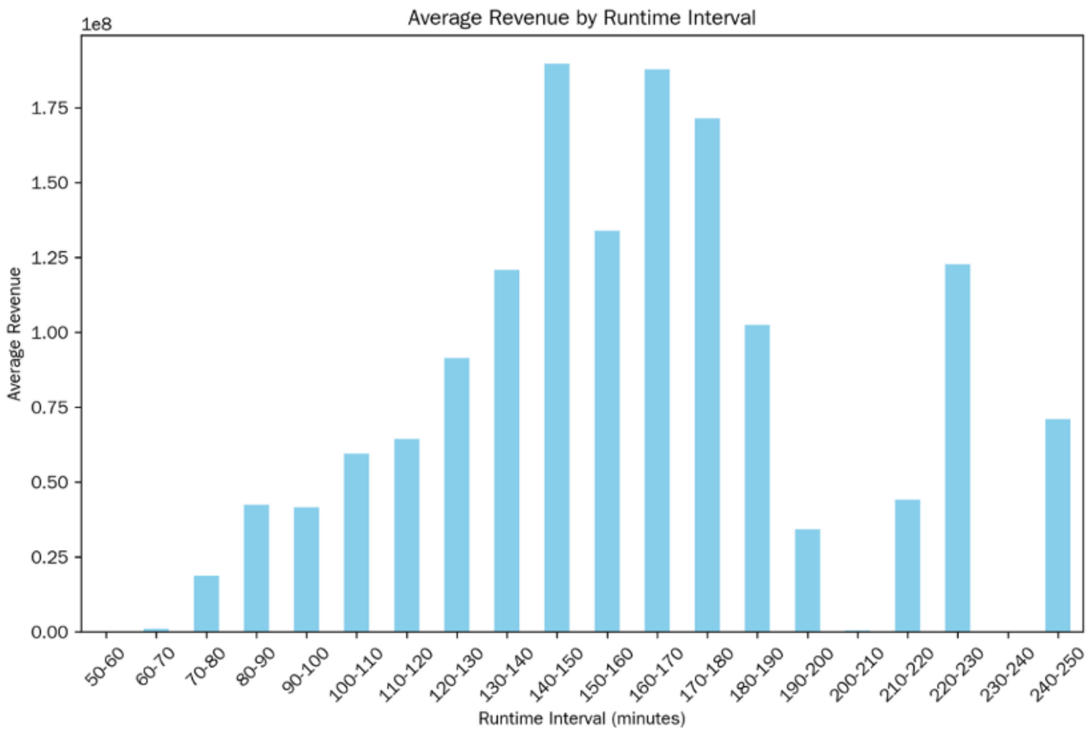
3.3 **Data Analysis And Visualization**

The data exploration phase integrated statistical summaries and visualizations to identify patterns that influence box office performance. Fundamental statistics, including mean, median, standard deviation, and others, provided essential insights into the distribution and variability of key variables, such as budget, runtime, and revenue. Visual representations further elucidated complex relationships, uncovering previously obscured features.

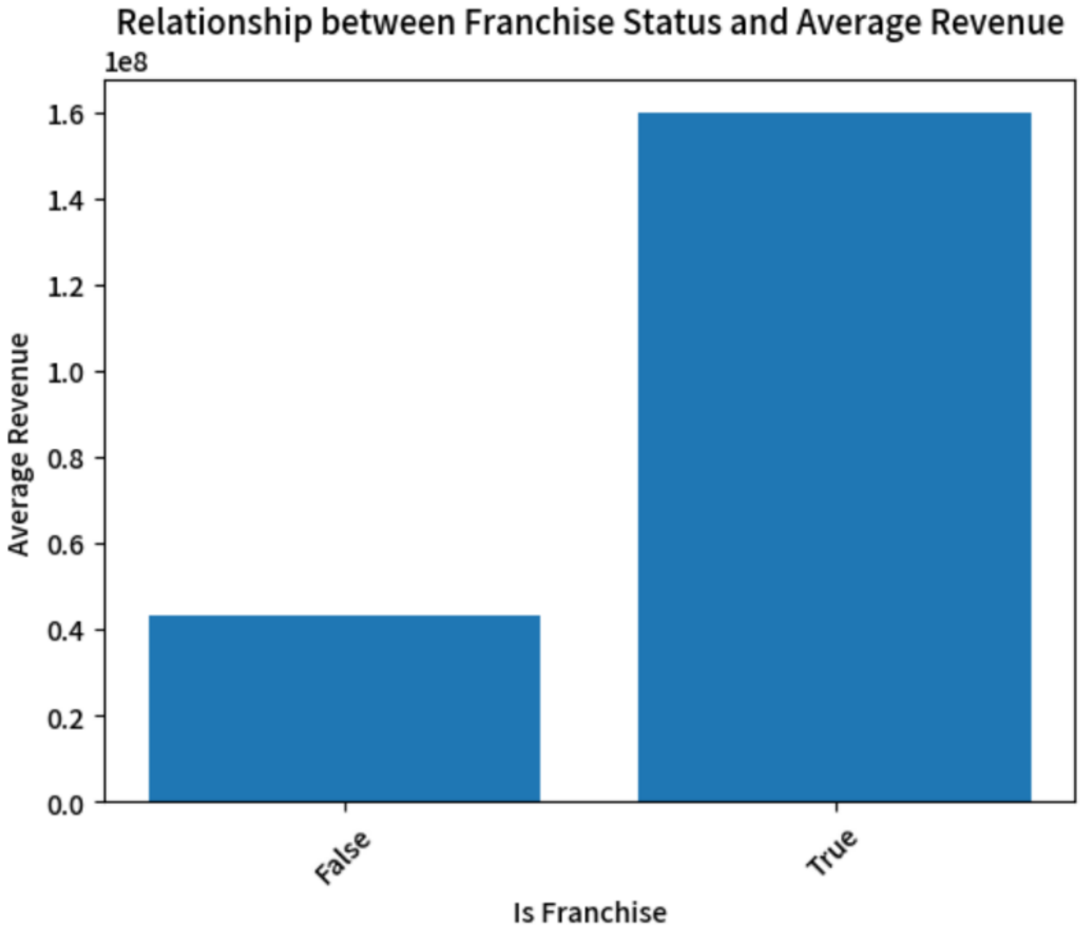
Budget-Revenue Relationship: The log-transformed scatter plot demonstrated a significant positive correlation (R^2=0.57), indicating that budget accounts for 57% of the variation in revenue.



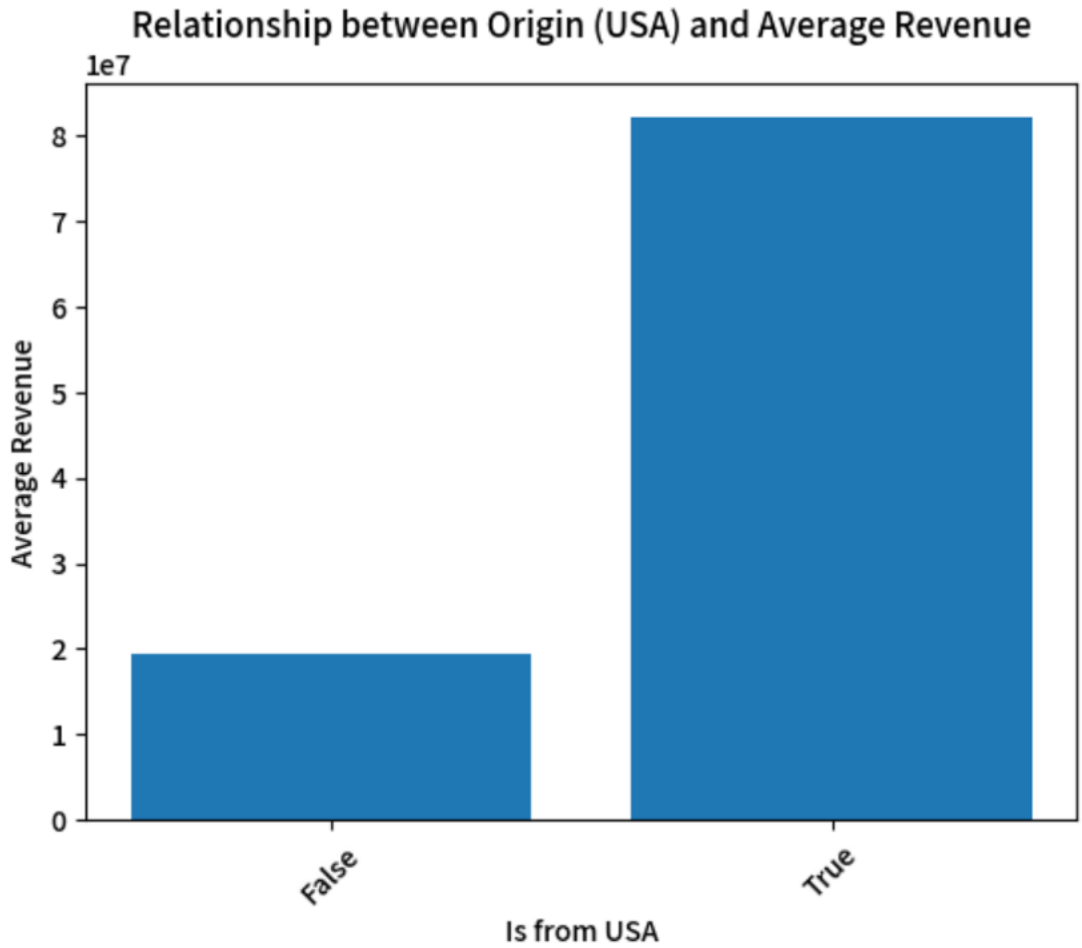
Optimal Runtime: Films lasting 140–180 minutes generated the highest revenue, while excessively short or long runtimes underperformed.



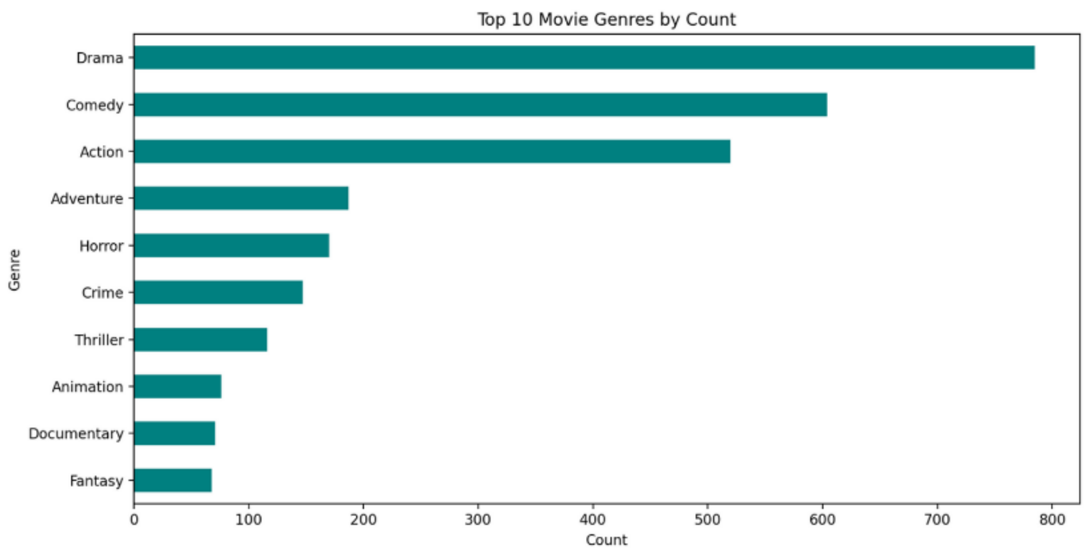
Franchise Advantage: Franchise films (e.g., *Marvel* or *Fast & Furious*) earned 1.5 times more than standalone movies, attributed to built-in fanbases.



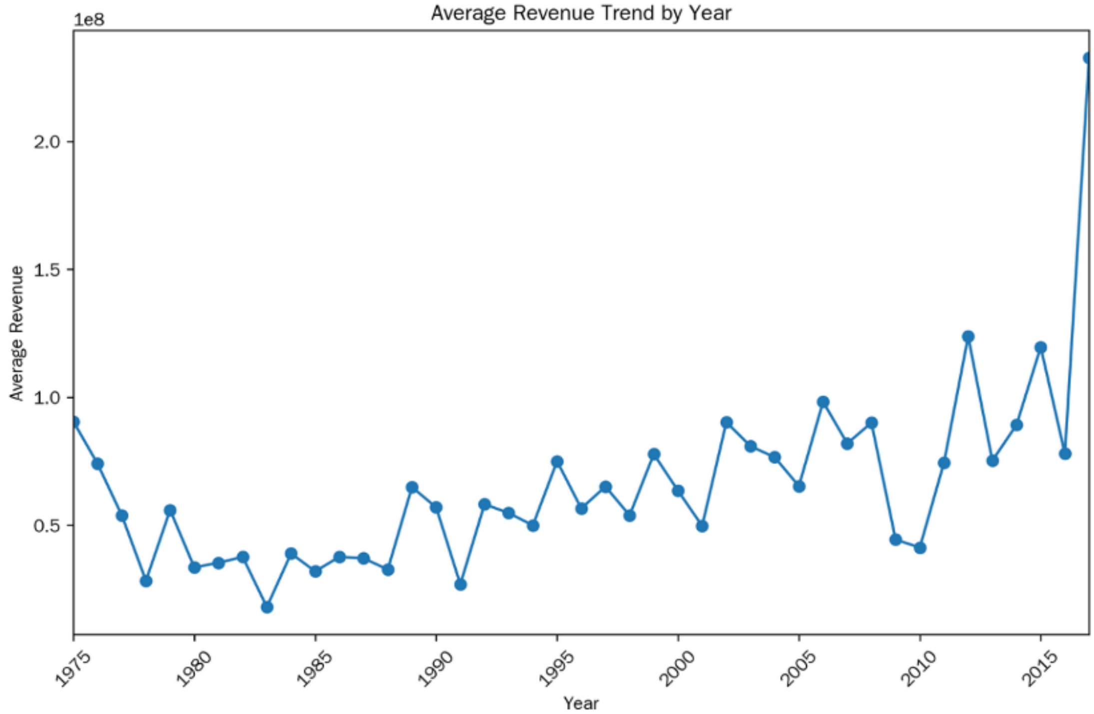
US Production Dominance: Hollywood's resources and global distribution networks contributed to stronger box office performance for US-produced films.



Genre Trends: Drama, comedy, and action dominated in quantity, though genre-specific revenue averages varied.



Temporal Growth: Post-2020, global revenue surged, likely driven by streaming expansion and emerging markets.



In conclusion, we selected seven key attributes from the twenty-one variables available in the database. These attributes include budget, optimal runtime (ranging from 140 to 180 minutes), franchise status, U.S. production, genre popularity, release year, and season. The excluded variables—namely, spoken language, poster, and company—were determined to lack consistency and generalizability. This selection can be quantitatively assessed and distinctly categorized, facilitating the development of a balanced and reliable prediction framework.

3.4 Model (Random Forest)

3.4.1 What Is It?

A Random Forest represents an ensemble learning methodology that constructs a multitude of decision trees throughout the training process. In the context of your project, the forest comprises 300 individual decision trees. Each tree undergoes training on a random subset of the training data utilizing replacement (referred to as bootstrapping). Furthermore, during the process of splitting nodes within each tree, only a random subset of features (specifically, three features as stated in max_features=3) is taken into consideration. This synthesis of bootstrapping and random feature selection is fundamental in ensuring the diversity among the trees, which is critical to enhancing the model's performance.

3.4.2 Why Did We Use It?

1. The Random Forest model was selected for several key justifications that render it appropriate for forecasting movie box office revenue:

2. It adeptly manages complex relationships within the data, encompassing both non-linear and linear patterns, which are likely to be present in factors influencing box office success.
3. It performs effectively with various types of data, accommodating both numerical and categorical attributes utilized in the dataset.
4. The ensemble approach, particularly involving bootstrapping and feature randomness, contributes to the mitigation of overfitting, thereby enhancing generalization on unseen data.
5. It exhibits robustness against noise within the data, indicating a diminished sensitivity to minor errors or outliers in comparison to certain alternative models.

3.4.3 How Does It Work?

This project utilized Random Forest to perform regression, aiming to predict a continuous value: movie box office revenue.

3.4.3.1 Hyperparameters

Several hyperparameters were established to configure the model's behavior:
n_estimators = 300: This parameter specifies the number of decision trees in the ensemble forest.

max_depth = 20: This parameter limits the maximum depth of each individual tree, thereby preventing the trees from becoming excessively complex and overfitting to noise present in the training data.

max_features = 3: This parameter determines the number of features that each tree randomly considers when seeking the optimal split at a node, which promotes diversity among the trees.

random_state = 42: This parameter sets a seed for the random number generator, ensuring that the bootstrapping process and the random feature selection remain consistent each time the code is executed, thus rendering the results reproducible.

3.4.3.2 Prediction on New Movies

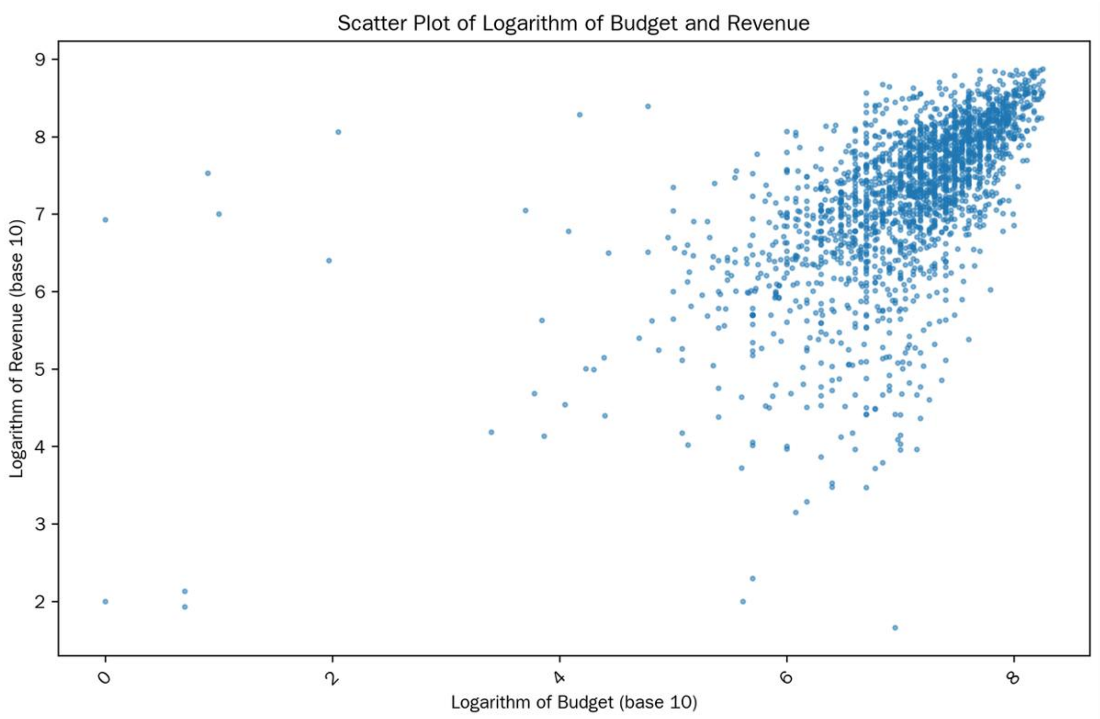
The internal mechanism of the model operates in the following manner to generate a prediction:

1. Each decision tree within the forest utilizes the feature values of the new movie as input data.

2. Commencing from the root node of the decision tree, it executes a sequence of binary decisions predicated on the feature values. For instance, at a specific node, it may assess whether the budget_log exceeds a designated threshold.
3. As a consequence of these determinations, the tree progresses through the nodes until it arrives at a leaf node. Each leaf node is associated with a predicted value for the log-transformed revenue.

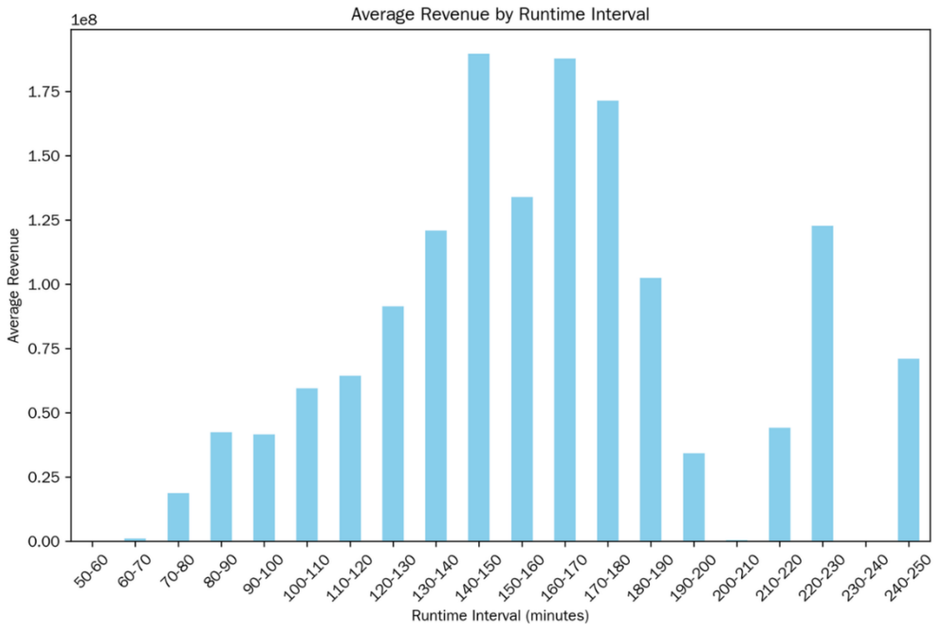
4. Results

4.1 Budget vs Revenue



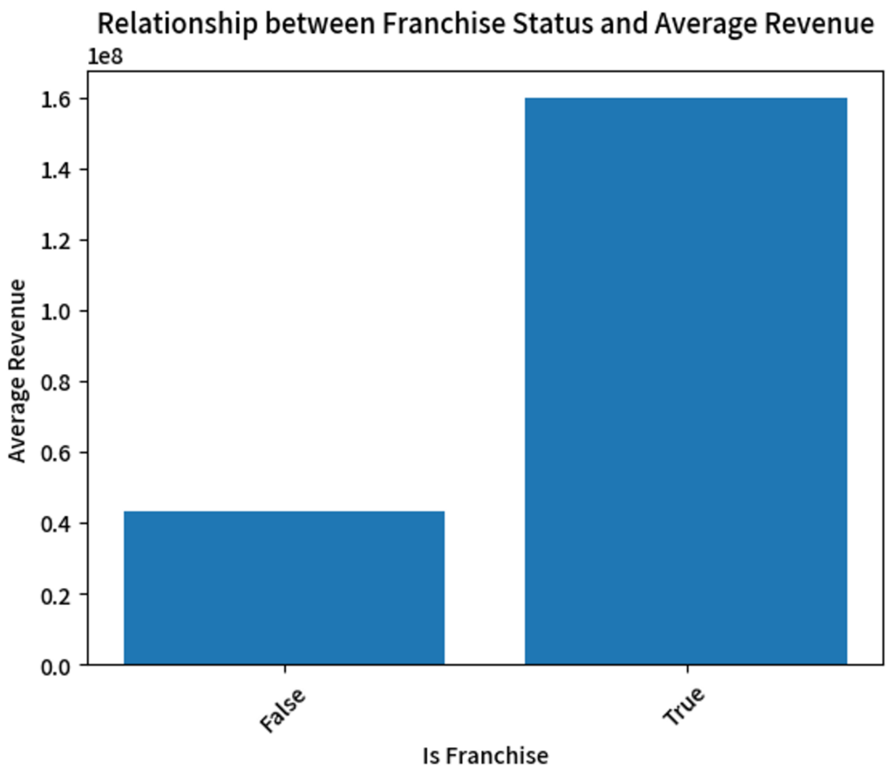
There is a positive correlation between budget and revenue, with an R^2 of 0.57.

4.2 Runtime vs Budget



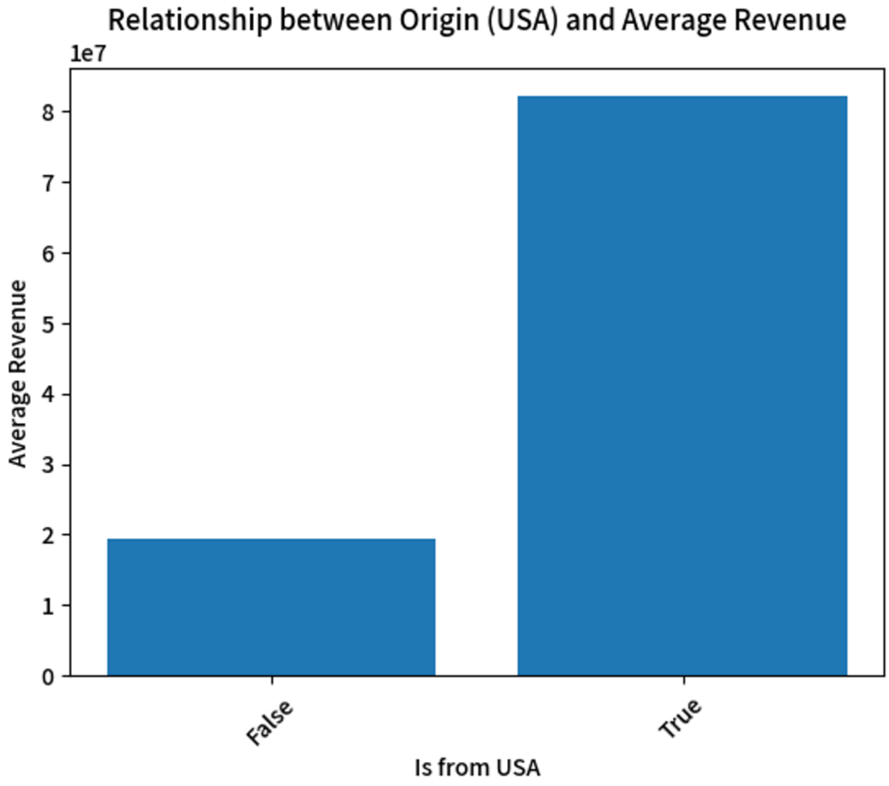
Movies with runtimes between 140-150 minutes and 160-170 minutes show the highest average revenue, both exceeding $180 million. Runtimes under 70 minutes and between 200-220 minutes generally have lower average revenues.

4.3 Franchise vs Revenue



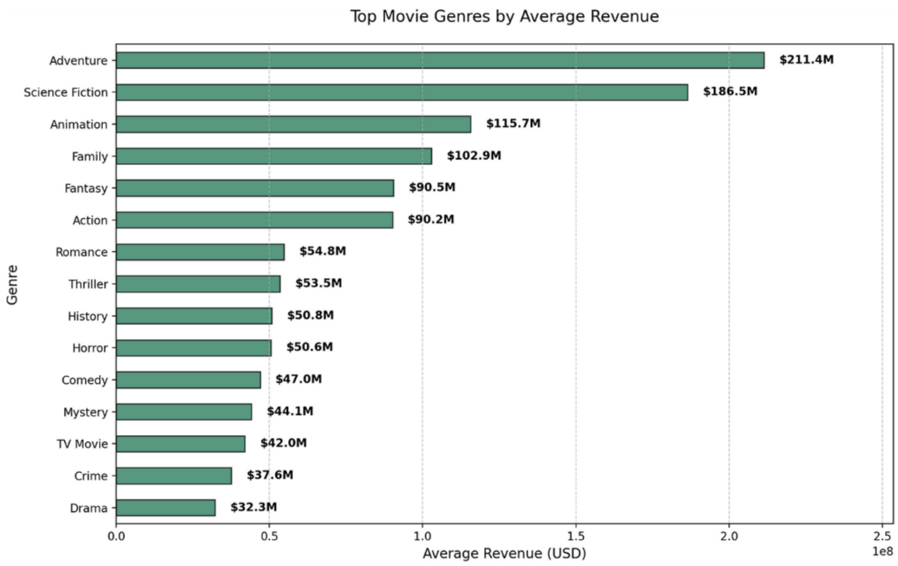
Movies that are part of a franchise ("True") have significantly higher average revenue (around $160 million) compared to movies that are not ("False") (around $40 million).

4.4 US Production vs Revenue



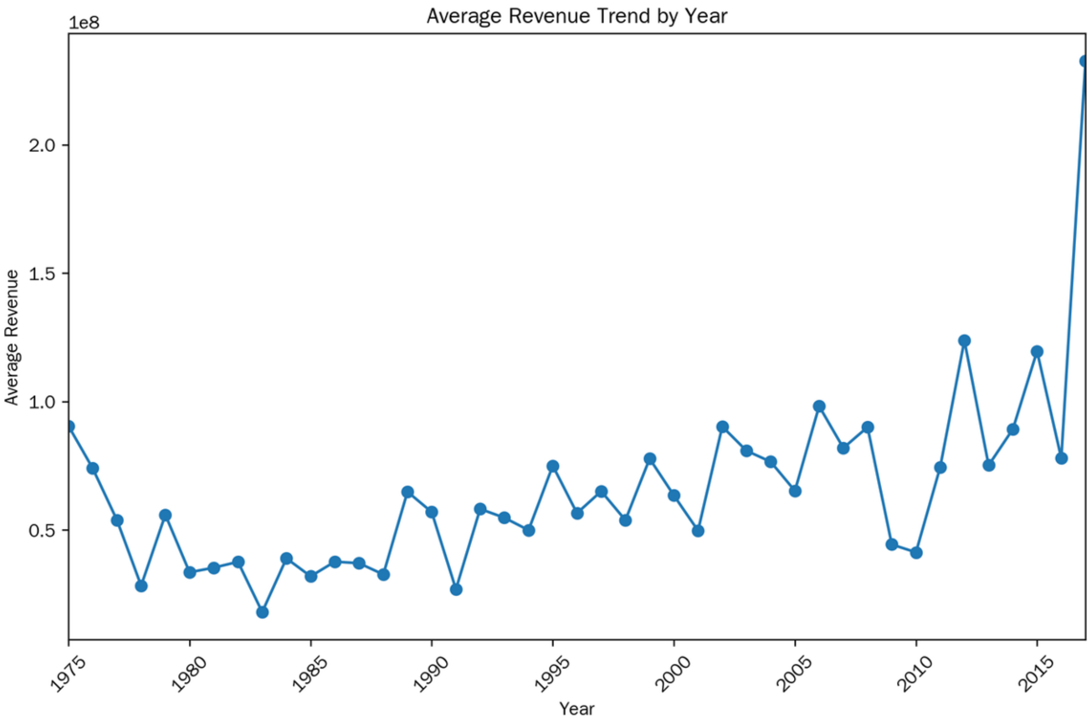
Movies originating from the USA ("True") have a much higher average revenue (over $80 million) than movies not from the USA ("False") (around $20 billion).

4.5 Genres vs Revenue



The top-performing genres by average revenue are Adventure ($211.4M), Science Fiction ($186.5M), and Animation ($115.7M). Drama has the lowest average revenue among the listed genres at $32.3M.

4.6 Year vs Revenue



Average movie revenue shows fluctuations over the years. There appears to be a general upward trend with significant peaks and dips. Notably, there is a sharp increase in average revenue in the most recent years shown (around 2016).

4.7 Random Forest Test Results

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Model Evaluation:

Simple Random Forest Performance:
- MAE: $45,302,895
- RMSE: $102,172,011
- R²: 0.61
- Accuracy (within 30% error margin): 53.88%
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The presented metrics indicate that, on average, the model's predictions deviated by approximately $45.3 million (MAE), with a more pronounced average error of roughly $102.2 million (RMSE), which imposes a greater penalty on larger discrepancies. The R^2 value of 0.61 indicates that the model accounts for 61% of the variance in box office revenue. Furthermore, the model successfully predicted the box office revenue within a 30% error margin approximately 53.88% of the time.

5. Findings and Improvements

5.1. Findings

The budget shows a strong correlation with revenue, with an R^2 value of 0.57, indicating that effective funding is crucial in film production. The genre_popularity feature reveals that adventure and science fiction genres typically generate the highest revenue, while crime and drama produce the lowest. This knowledge helps filmmakers and investors maximize profitability. Understanding the link between genre and revenue can guide investment and production choices. Focus on adventure and science fiction may yield higher returns, while caution is advisable with crime and drama due to their lower revenue potential. By applying these insights, filmmakers and investors can align projects with audience preferences and enhance financial outcomes.

5.2 Limitations and Improvements

Limitations: The study's findings are limited by data quality and scope. Key limitations include incomplete budget and revenue records, which may not reflect evolving audience preferences. The model assumes linear relationships after log-transformation, potentially missing complex interactions. Additionally, excluded variables like spoken language or poster effects might have unquantified predictive power. Potential Improvements: Enhancements could involve real-time social media sentiment analysis, macroeconomic indicators, or streaming performance data. Advanced modeling techniques like LightGBM or CatBoost may better capture nonlinear patterns. Expanding data diversity and refining genre metrics could improve generalizability. Future Research: Future studies could explore cross-regional market dynamics, streaming's impact on theatrical releases, or detailed marketing strategies. Investigating feature interactions may refine predictive accuracy further. For instance, dimensionality reduction techniques like PCA could merge related features, enhancing accuracy.

6. Conclusion

This project developed a Random Forest model aimed at predicting box office revenue for movies through comprehensive data analysis and feature engineering. Key factors such as budget, franchise status, runtime, and genre popularity were identified as significant predictors, with budget exhibiting the strongest correlation to revenue (R^2=0.57). The model adeptly managed complex relationships within the data while minimizing overfitting, thereby yielding reliable predictions. Through rigorous data cleaning, transformation, and visualization, substantial patterns regarding movie performance were uncovered. The results present valuable insights for industry stakeholders, illustrating how data-driven strategies can enhance the understanding of box office success. This work establishes a practical framework for revenue prediction within the film industry.

7. Appendix

The visual results of the predicted box office are attached to this file (movie_predictions.html).

References

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